

MOHAMMAD TAAHA ASHRAF

Software Engineer | AI Engineering & Backend Systems

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EDUCATION

Jamia Hamdard University

New Delhi, India

B.Tech. in Computer Science & Engineering

Expected May 2027

Coursework: Data Structures & Algorithms (C++), Object-Oriented Programming, Database Management Systems, System Design

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Frontend: React, Next.js, Tailwind CSS, Base UI, HTML5, CSS3

Backend & APIs: FastAPI, Node.js, Express.js, Celery, REST APIs, WebSockets, Socket.io, WebRTC

Databases: PostgreSQL, SQLAlchemy 2.x (async), MySQL, MongoDB, Redis, Prisma

AI Engineering: Gemini API, Pydantic, Structured Output, LLM Integration

DevOps & Testing: Docker, GitHub Actions, Nginx, Turborepo, pnpm, Git, GitHub, GitLab, Vercel, Render, Jest, Supertest, Postman, Joi

PROJECTS

Musafir Trips | Full-Stack Travel Agency Platform

[Live Demo] | [GitHub]

Next.js 14, TypeScript, Prisma 7, PostgreSQL (Neon), Auth.js v5, Tailwind CSS, Cloudinary

- Built a CMS-driven travel platform with an admin dashboard covering tours, blogs, hero slides, enquiries, awards, and site-wide settings, deployed on Vercel.
- Implemented JWT authentication with Google OAuth and credential login via Auth.js v5, gated by Edge middleware for route-level access control.
- Used Next.js Server Actions with Prisma 7 for all database mutations, triggering automatic ISR revalidation on every content update.
- Integrated Cloudinary image uploads and a Tiptap rich-text editor across admin content workflows.

exac.draw | Real-Time Collaborative Whiteboard

[Live Demo] | [GitHub]

Next.js 15, Turborepo, WebSockets, PostgreSQL, Tailwind CSS, pnpm

- Designed a Turborepo monorepo with shared packages between the frontend and backend, streamlining cross-environment code sharing and tooling.
- Engineered real-time canvas synchronization over WebSockets, enabling low-latency, simultaneous multi-user editing.
- Built a custom canvas rendering engine for geometric shapes, backed by persistent whiteboard state in PostgreSQL.

NitPick | AI-Powered Code Review Platform

[GitHub]

Python, FastAPI, SQLAlchemy 2.x (async), PostgreSQL, Celery, Redis, Gemini API, Pydantic

- Designing an AI-powered code review platform with two entry points, direct code upload and GitHub pull request review, that converge into a single review pipeline.
- Architecting an asynchronous review pipeline with FastAPI, Celery, and Redis to decouple API and GitHub webhook requests from long-running Gemini-based processing.
- Building structured-output validation with Pydantic and semantic checks to validate Gemini-generated findings before persisting them to PostgreSQL through SQLAlchemy's async ORM.

kidsportal | Interactive Learning Platform

[GitLab]

FastAPI, Python, React (Vite), InkJS, MySQL

- Collaborated in a 3-person team to build an administrative platform for educational content, analytics, and user progress.
- Developed a custom "Stitcher" compiler that converts database-stored story fragments into InkJS-compatible bytecode for branching narratives.
- Built a FastAPI backend to manage real-time interactive story states and concurrent student session data.

OPEN SOURCE CONTRIBUTIONS

CropChain (Aperture 2.0) | Open Source Contributor

[GitHub]

Node.js, Express, Joi, Jest, Supertest, JavaScript

- Secured a previously unprotected backend route by designing Joi validation schemas and a centralized request-validation middleware.
- Built an automated integration test suite with Jest and Supertest to validate API behavior and error-handling paths.

LEADERSHIP & ACTIVITIES

Tezos Society, Jamia Hamdard | Tech Lead (Member since 2024)

2025–Present

- Lead a team of student developers running technical initiatives and events for the university's Web3 society.
- Launched DSA-focused practice sessions for the tech team to help new members build stronger programming fundamentals.